

Cyclone Detection and Intensity Estimation using Satellite Images

Shrikant Hamand Anjali Ingle Dhananjay Kokate

MIS: 112103048

MIS: 112103051

MIS: 112103073

Abstract

Tropical cyclones pose a serious risk to infrastructure and human life, particularly in coastal regions. Traditional cyclone detection and intensity estimation methods, such as the Dvorak approach, rely on manual interpretation of satellite images, which is frequently subjective, time-consuming, and less responsive. Motivated by advances in artificial intelligence and remote sensing, our study presents a deep learning-based framework that uses satellite imagery to automatically detect cyclones and predict their intensity. Our system integrates multi-modal data from INSAT-3D satellite imagery, focusing on thermal infrared channels to capture temperature anomalies and cloud structures.

We use optical flow and super-resolution techniques to improve low-resolution images, and we use convolutional neural networks (CNNs) to estimate intensity. A real-time processing pipeline that facilitates quick predictions and visual feedback via a web-based interface is made possible by our lightweight architectures. Using historical cyclone datasets covering events from 2013 to 2021, the system was trained and assessed. It achieved over High accuracy and a low Root Mean Squared Error (RMSE) in wind speed prediction.

Additionally, users can upload satellite photos for cyclone analysis and view dynamic wind patterns on a live weather dashboard. Our model's ability to generalize across regions, produce timely outputs, and withstand data heterogeneity all demonstrate its practical value in disaster preparedness and emergency response. Our research uses deep learning's revolutionary potential in operational meteorology to provide a scalable, accurate, and effective method of reducing the effects of tropical cyclones.

Keywords: Deep learning, Convolutional Neural Network, Cyclone detection, Wind speed estimation, Satellite imagery, Disaster management, INSAT-3D, Optical flow, Super-resolution.